

TrES-5b

R-1883

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_190804 · created 2026-08-03T12:02:53+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458699.7462 +/- 0.0029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.053
Transit depth (Rp/Rs)^2	3.42 +/- 1.98 [%]
Orbital Inclination (inc)	89.31 +/- 2.91
Airmass coefficient 1 (a1)	10.2 +/- 6.68
Airmass coefficient 2 (a2)	-2.05 +/- 1.58
Scatter in the residuals of the lightcurve fit is	60.06 %
Best Comparison Star	#1 - [302, 370]
Optimal Aperture	0.8
Optimal Annulus	3.45
Transit Duration (day)	0.0872 +/- 0.0079

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.185 ± 0.053 vs 0.142 (deviation 0.81σ)
Duration fit vs published	0.0872 d vs 0.0758 d (deviation 1.44σ)
Mid-time O–C	-4.1 min
Depth SNR	3.5
Residual scatter	60.06 %

Light curve

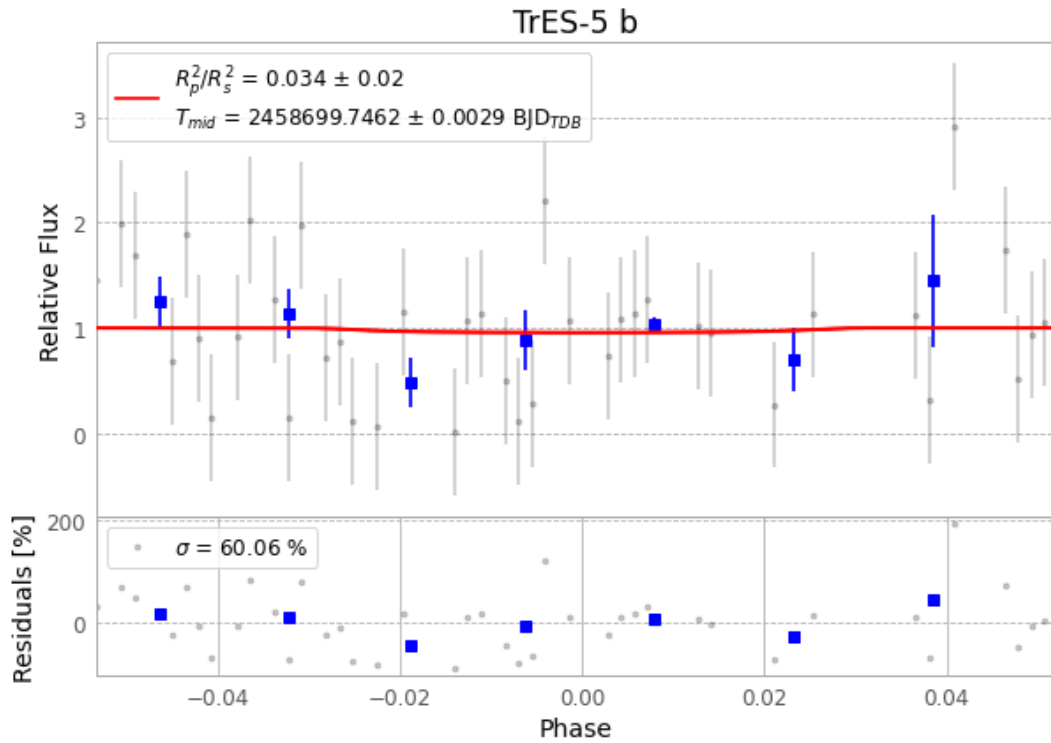


Figure 1 — FinalLightCurve_TrES-5 b_2019-08-03.png (SHA-256 39af39dc38fe3210...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [492, 151], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 01581795cf98cf7a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), TRES-5190804035712.FITS → TRES-5190804074512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-190804.log (8fa59d9c4b18...), FinalLightCurve_TrES-5 b_2019-08-03.png (39af39dc38fe...), FinalLightCurve_TrES-5 b_2019-08-03.csv (349b94931c42...)

Compute resources

Wall time 220.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 12:02Z.